Low-Code Experiment

Configuring User Management and Role-Based Access Control (RBAC) in a Zoho Creator Application

Sachitha S
241501171
AIML

Abstract
This experiment studies user management and role-based access control in the HOTEL BOOKING application. The User Permissions page of Settings, with its Permissions, Roles and Data Sharing tabs, was explored, a new permission set was configured module by module for the Room Booking Form and Booking Dashboard, and the live application was opened to see the menu available to the administrator.

1. Aim
To configure user management and role-based access control (RBAC) so that each user sees only the forms, reports and pages that suit their role.

2. Concepts
Permissions. The Permissions tab of the Users page lists named permission sets. Two defaults exist: Read, with read permission for all components, and Write, with write permission for all components. New sets are added with the Add Permission button.

Module-level control. In the Add Permission Set dialog, a table lists every module (forms and pages) with the rights Access, View (the report shown), Edit, Delete and More. A user with the set can use only the modules and actions ticked in it.

Roles and data sharing. The Roles tab defines a role hierarchy and the Data Sharing tab defines the visibility of records for each user, in addition to the permission set.

User management. The application is shared with users by email address using the Share button, and each user is given a permission set. The administrator can change it later.

3. Procedure
1. Open the HOTEL BOOKING application, click Settings (gear icon) and then User Permissions.
2. Observe the three tabs of the Users page: Permissions, Roles and Data Sharing, and the default permission sets Read and Write.
3. Click Add Permission to open the Add Permission Set dialog and scroll to the Set Permissions table.
4. For the Room Booking Form, tick Access and choose the report shown in the View column. Leave Edit, Delete and More unticked so that users with this set can only work with bookings, not change or delete them. Tick Access for the Booking Dashboard. Save the set.
5. Open the live application with Access Development Live and review the menu available to the administrator: Room Booking Form, Room Booking Report, Upcoming Bookings Report, Bookings By Room Type and Booking Dashboard.
6. To grant the set to a person, use the Share button, enter the email address and choose the permission set.

4. Implementation Screenshots (HOTEL BOOKING application)

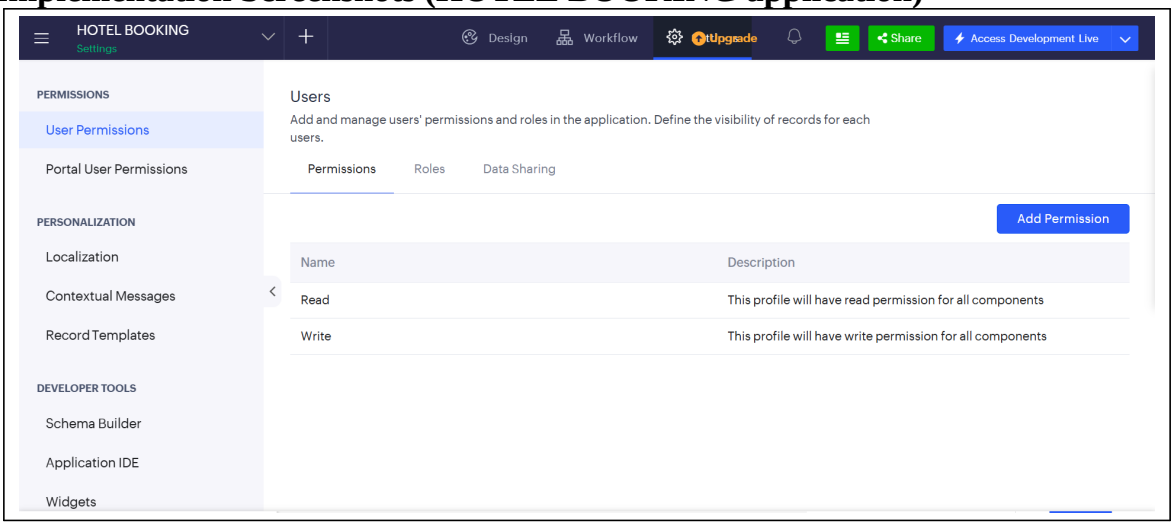


Figure 1: Users page of Settings with the Permissions, Roles and Data Sharing tabs and the Read and Write sets

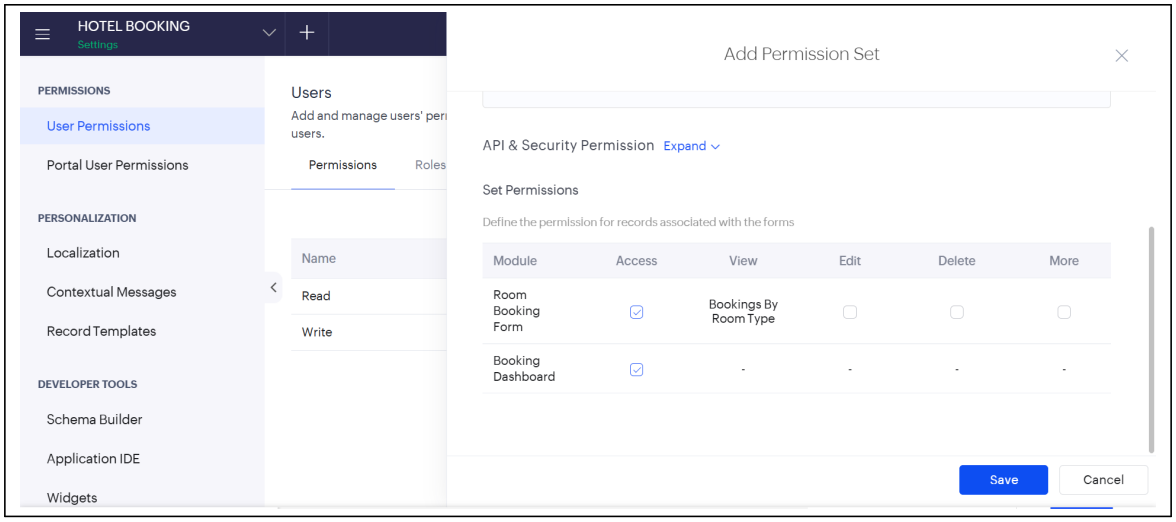


Figure 2: Add Permission Set dialog: Set Permissions table for the Room Booking Form and Booking Dashboard

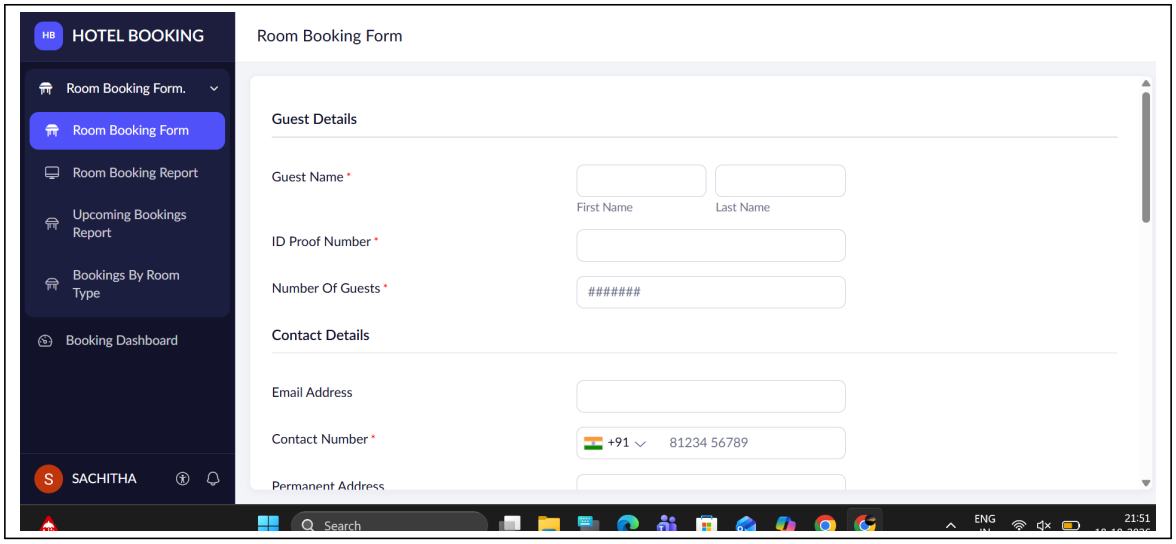


Figure 3: Live application as the administrator: forms, reports and the Booking Dashboard in the menu

5. Expected Output

The Users page lists the available permission sets, a new set can control Access, View, Edit, Delete and More for each form and page, and the live application lists all forms, reports and the dashboard for the

administrator. Users given a restricted set see only the modules ticked in it.

6. Result

The procedure above was prepared for the HOTEL BOOKING application. The observed results are recorded through the screenshots in the figures above. Where a feature is restricted because the free trial has expired, the message shown by Zoho Creator is included as the screenshot and treated as the observed result.

7. Learning Outcome

Learnt how permission sets, roles and data sharing implement role-based access control in Zoho Creator, how to apply the principle of least privilege module by module, and how users are granted access through the Share option.